

A complex function involving 10 Boolean variables can be defined as:

$$f(x_1, x_2, \dots, x_{10}) = (x_1 \oplus x_2) \wedge (x_3 \vee x_4) \oplus (x_5 \wedge x_6) \vee (x_7 \oplus x_8) \wedge (x_9 \oplus x_{10})$$

where:

- $\wedge$ represents the AND operation.
- $\vee$ represents the OR operation.
- $\oplus$ represents the XOR operation.

This function combines multiple logical operations to produce a complex decision boundary.

1. Construct a decision table listing all possible input combinations and their corresponding outputs for the given Boolean function.
2. Design an Artificial Neural Network (ANN) model with two hidden layers, each containing five nodes.
3. Develop a detailed mathematical representation of the ANN, including weight transformations, activation functions, and forward propagation equations.
4. Implement the ANN model from scratch in Python, incorporating weight initialization, forward propagation, backpropagation, and optimization techniques.
5. Apply GD, SDG and ADAM optimizers and plot the convergence rate.
6. Performance evaluation (accuracy, confusion matrix, precision, recall, F1-score)
7. Convergence plot and performance metrics for different combinations of learning rate and number of epochs.

1. How does the convergence rate of GD, SGD, and AMDA change with different learning rates?
2. What is the impact of varying the number of epochs on the accuracy and loss reduction of each optimizer?
3. Which optimizer achieves the lowest loss for a given learning rate and epoch combination?
4. How does the stability of the optimization process vary among GD, SGD, and AMDA for different learning rates?
5. What is the effect of increasing the number of epochs on overfitting for each optimizer?
6. How does the variance in updates differ among GD, SGD, and AMDA across different learning rates?
7. Which optimizer performs best in terms of computational efficiency for different epoch settings?
8. What is the effect of learning rate decay on the convergence of these optimizers?
9. How does the choice of optimizer impact model generalization when evaluated on test data?
10. How does the learning rate influence the speed of convergence for each optimizer?
11. What are the optimal learning rate and epoch combinations for minimizing loss in each optimizer?
12. How does the variance in gradient updates affect the stability of GD, SGD, and AMDA?
13. Which optimizer demonstrates the best trade-off between speed and accuracy?
14. How does the initial learning rate affect the final accuracy achieved by each optimizer?
15. What is the impact of different batch sizes in combination with learning rate and epochs for SGD and AMDA?
16. How does each optimizer handle noisy gradients in the training process?
17. Which optimizer is most robust to changes in hyperparameters (learning rate and epochs)?
18. How does the choice of optimizer influence model convergence in high-dimensional optimization problems?

19. How does the computational time per epoch vary across GD, SGD, and AMDA?
 20. How do the optimizers compare in handling sparse datasets when varying learning rate and epochs?
 21. What is the role of momentum in AMDA, and how does it influence convergence compared to GD and SGD?
 22. How do these optimizers perform in different loss landscapes (convex vs. non-convex)?
 23. What is the effect of different initialization strategies on the convergence behavior of each optimizer?
 24. How do adaptive learning rate strategies affect the performance of AMDA in comparison to GD and SGD?
 25. How does the choice of optimizer affect the sensitivity of model weights to small perturbations in training data?
 26. What is the impact of early stopping on each optimizer when trained with different learning rates and epochs?
 27. How do the optimizers compare in terms of energy efficiency when run on hardware like GPUs or TPUs?
 28. Which optimizer demonstrates the most consistent performance across multiple datasets?
 29. How do the optimizers handle vanishing and exploding gradient problems with varying learning rates and epochs?
- ### Optimization Behavior & Convergence
30. How does the rate of convergence differ for GD, SGD, and AMDA in different optimization scenarios?
 31. What is the relationship between step size (learning rate) and oscillations in the loss function for each optimizer?
 32. How does the learning rate affect the tendency of each optimizer to get stuck in local minima?
 33. What are the differences in convergence speed for convex vs. non-convex loss functions?
 34. How does the presence of saddle points affect the convergence of each optimizer?
 35. What is the impact of learning rate warm-up strategies on the performance of each optimizer?

Generalization & Overfitting

36. How does increasing the number of epochs influence overfitting for each optimizer?
37. Which optimizer generalizes best to unseen test data when trained with different learning rate schedules?
38. How does model performance vary when regularization techniques (L1, L2, dropout) are used with each optimizer?
39. How does early stopping based on validation loss impact the performance of different optimizers?
Hyperparameter Sensitivity & Adaptability
40. How does the choice of an adaptive learning rate (such as Adam or RM-Sprop) compare to AMDA in performance?
41. How does fine-tuning the learning rate schedule affect performance in each optimizer?
42. What is the effect of exponential decay of the learning rate on the stability of each optimizer?
43. Which optimizer is the most robust to hyperparameter variations across different datasets?
44. How does batch normalization impact the effectiveness of each optimizer?
Computational Efficiency & Scalability
45. How does the training time per epoch compare for GD, SGD, and AMDA on different hardware architectures (CPU vs. GPU)?
46. Which optimizer achieves the best trade-off between computational efficiency and accuracy?
47. How does parallelization (e.g., distributed training) affect the performance of each optimizer?
48. What is the memory footprint of each optimizer when scaling to larger datasets?
49. How does the training efficiency change when using mixed-precision floating-point operations (FP16 vs. FP32)? ### Robustness to Data Variability
50. How does data imbalance affect the optimization process for GD, SGD, and AMDA?
51. How does the performance of each optimizer vary with noisy or corrupted labels in the dataset?
52. Which optimizer is more resistant to adversarial examples when trained with different learning rates and epochs?

- 53. How do the optimizers perform on datasets with different feature distributions (e.g., skewed, heavy-tailed, uniform)?
- 54. How does dataset augmentation influence the performance of each optimizer?
Handling of Specific Training Challenges
- 55. Which optimizer is most effective in mitigating vanishing gradients in deep networks?
- 56. How does each optimizer handle high-dimensional feature spaces with sparse inputs?
- 57. How does each optimizer handle sharp vs. flat minima in the loss landscape?
- 58. Which optimizer is better at escaping poor local minima in deep learning models?
- 59. How does the presence of outliers in the dataset affect the stability and convergence of different optimizers?